GLUEING SHEAVES OF SETS ALONG AN OPEN COVERING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma lemma-extend-off-basis (canonical tag 009N): a sheaf on a base of the topology extends uniquely to a sheaf on the space. invoked only inside the author's supplementary second proof (sketch), proof lines 5460-5465. Not load-bearing for the counted first proof; navigation stub for the sketch paragraph.

Source reference. Original lemma 009N: sources/stacks-chow/tags.txt, source lines 359–359. Invoked only in the author's supplementary second proof (sketch) at proof lines 5460-5465; navigation stub, not load-bearing for the counted first proof.

Lemma 0.2. *Let X be a topological space. Let $X = \bigcup_{i \in I} U_i$ be an open covering. Given any glueing data $(\mathcal{F}_i, \varphi_{ij})$ for sheaves of sets with respect to the covering $X = \bigcup U_i$ there exists a sheaf of sets $\mathcal{F}$ on X together with isomorphisms*

$$\varphi_i : \mathcal{F}|_{U_i} \to \mathcal{F}_i$$

such that the diagrams

$$\begin{array}{ccc} \mathcal{F}|_{U_i \cap U_j} & \xrightarrow{\varphi_i} & \mathcal{F}_i|_{U_i \cap U_j} \\ {\scriptstyle id}\downarrow & & \downarrow{\scriptstyle \varphi_{ij}} \\ \mathcal{F}|_{U_i \cap U_j} & \xrightarrow{\varphi_j} & \mathcal{F}_j|_{U_i \cap U_j} \end{array}$$

are commutative.

Proof. First proof. In this proof we give a formula for the set of sections of $\mathcal{F}$ over an open $W \subset X$. Namely, we define

$$\mathcal{F}(W) = \{(s_i)_{i \in I} \mid s_i \in \mathcal{F}_i(W \cap U_i), \varphi_{ij}(s_i|_{W \cap U_i \cap U_j}) = s_j|_{W \cap U_i \cap U_j}\}.$$

Restriction mappings for $W' \subset W$ are defined by the restricting each of the s_i to $W' \cap U_i$. The sheaf condition for $\mathcal{F}$ follows immediately from the sheaf condition for each of the $\mathcal{F}_i$.

We still have to prove that $\mathcal{F}|_{U_i}$ maps isomorphically to $\mathcal{F}_i$. Let $W \subset U_i$. In this case the condition in the definition of $\mathcal{F}(W)$ implies that $s_j = \varphi_{ij}(s_i|_{W \cap U_j})$. And the commutativity of the diagrams in the definition of a glueing data assures that

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we may start with *any* section $s \in \mathcal{F}_i(W)$ and obtain a compatible collection by setting $s_i = s$ and $s_j = \varphi_{ij}(s_i|_{W \cap U_j})$.

Second proof (sketch). Let $\mathcal{B}$ be the set of opens $U \subset X$ such that $U \subset U_i$ for some $i \in I$. Then $\mathcal{B}$ is a base for the topology on X. For $U \in \mathcal{B}$ we pick $i \in I$ with $U \subset U_i$ and we set $\mathcal{F}(U) = \mathcal{F}_i(U)$. Using the isomorphisms φ_{ij} we see that this prescription is "independent of the choice of i". Using the restriction mappings of $\mathcal{F}_i$ we find that $\mathcal{F}$ is a sheaf on $\mathcal{B}$. Finally, use Lemma 009N to extend $\mathcal{F}$ to a unique sheaf $\mathcal{F}$ on X. $\square$